problem_2

(a) Let x^* be the optimal solution to P^*

$\hat{x}$ be the optimal solution to $\hat{P}$

Both x^* and $\hat{x}$ satisfy the constraints $(Ax=b)$

Thus, for P^*, $\hat{x}$ is a feasible solution, so $c^T x^* \le c^T \hat{x}$

that is $-c^T(\hat{x}-x^*) \le 0$ ..... ①

Similarly, for $\hat{P}$, x^* is a feasible solution, $\hat{c}^T\hat{x} \le \hat{c}^T x^*$

that is $\hat{c}^T(\hat{x}-x^*) \le 0$ ..... ②

①+②: $(\hat{c}^T - c^T)(\hat{x}-x^*) \le 0$. Q.E.D.

(b) Let x^* be the optimal solution to P^*

$\tilde{x}$ be the optimal solution to $\tilde{P}$

By Theorem "Strong Duality", we have

$\pi^{*T} b = c^T x^*$ .... ①

$\tilde{\pi}^T \tilde{b} = c^T \tilde{x}$ .... ②

when let π^* be the optimal solution to D^*

$\tilde{\pi}$ be the optimal solution to $\tilde{D}$

The constraints of D^* and $\tilde{D}$ are same: $A^T\pi \le c$

Thus, π^* is a feasible solution to $\tilde{D}$. we have

$\pi^{*T}\tilde{b} \le \tilde{\pi}^T\tilde{b}$ .... ③ , when the object is maximizing $\tilde{b}^T\pi$

$-$①+②+③: $\pi^{*T}(\tilde{b}-b) \le c^T(\tilde{x}-x^*)$. Q.E.D.